JESÚS HUERTAS

jesus.huertas@ib.edu.ar 0000-0001-8230-9543 J.Huertas.2
Instituto Balseiro, San Carlos de Bariloche, 8400, Argentina, (+34) 648 583 539

POSITIONS

2025 – 2027 » **Simons-Balseiro Postdoctoral Fellow, Instituto Balseiro**, Bariloche, Argentina
2026 » Parental leave (4 months)

EDUCATION

Feb-May 2024 » **Fellow, Harvard University**, Cambridge, MA, USA
2021 – 2025 » **Ph.D., Instituto de Física Teórica UAM-CSIC**, Madrid, Spain
Advisor: Prof. Ángel Uranga
2020 – 2021 » **M.Sc. Theoretical Physics, Universidad Autónoma de Madrid**, Madrid, Spain
2016 – 2020 » **B.Sc. Physics, Universidad Complutense de Madrid**, Madrid, Spain

FELLOWSHIPS AND AWARDS

2025 » Simons-Balseiro Postdoctoral Fellowship
2023 » *Yo investigo, yo soy CSIC* outreach award (€3,000), CSIC
2021 – 2025 » *Contrato FPU*, PhD fellowship granted by the Spanish Government
2020 » *Premio Extraordinario de Grado*, award for the best academic record in the B.Sc.
2019 – 2020 » *Beca de colaboración*, research fellowship at Universidad Complutense granted by the Spanish Government

PUBLICATIONS

JOURNAL ARTICLES

1. H. Geng, J. Huertas, A. Karch, L. Randall, and D. Thomas, "Wet Hair: Global Symmetries in Entanglement Islands," Dec. 2025. arXiv: 2512.11025 [hep-th].
2. E. Anastasi, R. Angius, J. Huertas, A. M. Uranga, and C. Wang, "Relative quantum gravity: localized gravity and the swampland," *JHEP*, vol. 08, p. 107, 2025. DOI: 10.1007/JHEP08(2025)107. arXiv: 2501.03310 [hep-th].
3. J. Huertas and A. M. Uranga, "End of the world brane dynamics in holographic 4d $\mathcal{N}$ = 4 SU(N) with 3d $\mathcal{N}$ = 2 boundary conditions," *JHEP*, vol. 01, p. 002, 2025. DOI: 10.1007/JHEP01(2025)002. arXiv: 2410.05368 [hep-th].
4. I. García Etxebarria, J. Huertas, and A. M. Uranga, "SymTFT Fans: The Symmetry Theory of 4d N=4 Super Yang-Mills on spaces with boundaries," *JHEP*, vol. 03, p. 179, 2025. DOI: 10.1007/JHEP03(2025)179. arXiv: 2409.02156 [hep-th].
5. J. J. Blanco-Pillado, J. R. Espinosa, J. Huertas, and K. Sousa, "Bubbles of nothing: the tunneling potential approach," *JCAP*, vol. 03, p. 029, 2024. DOI: 10.1088/1475-7516/2024/03/029. arXiv: 2312.00133 [hep-th].
6. J. J. Blanco-Pillado, J. R. Espinosa, J. Huertas, and K. Sousa, "Tunneling potentials to nothing," *Phys. Rev. D*, vol. 109, no. 8, p. 084 057, 2024. DOI: 10.1103/PhysRevD.109.084057. arXiv: 2311.18821 [hep-th].

7. J. Huertas and A. M. Uranga, "Aspects of dynamical cobordism in AdS/CFT," *JHEP*, vol. 08, p. 140, 2023. DOI: 10.1007/JHEP08(2023)140. arXiv: 2306.07335 [hep-th].

8. R. Angius, J. Huertas, and A. M. Uranga, "Small black hole explosions," *JHEP*, vol. 06, p. 070, 2023. DOI: 10.1007/JHEP06(2023)070. arXiv: 2303.15903 [hep-th].

9. R. Angius, J. Calderón-Infante, M. Delgado, J. Huertas, and A. M. Uranga, "At the end of the world: Local Dynamical Cobordism," *JHEP*, vol. 06, p. 142, 2022. DOI: 10.1007/JHEP06(2022)142. arXiv: 2203.11240 [hep-th].

10. J. R. Espinosa, J.-F. Fortin, and J. Huertas, "Exactly solvable vacuum decays with gravity," *Phys. Rev. D*, vol. 104, no. 6, p. 065 007, 2021. DOI: 10.1103/PhysRevD.104.065007. arXiv: 2106.15505 [hep-th].

11. J. R. Espinosa and J. Huertas, "Pseudo-bounces vs. new instantons," *JCAP*, vol. 12, no. 12, p. 029, 2021. DOI: 10.1088/1475-7516/2021/12/029. arXiv: 2106.04541 [hep-th].

WORK IN PROGRESS

- DGP gravity from string theory, with E. Anastasi, R. Angius, A. M. Uranga, C. Wang (near completion)
- Islands and gravity localization, with M. Riojas
- Euclidean wormholes and the imaginary distance bound from tunneling potentials, with J. R. Espinosa
- Line operator SymTFTs from the Coulomb branch, with P. C. Argyres, A. Banerjee, A. Bourget, J. F. Grimminger, M. Lotito, S. Thakur, M. Weaver
- Brane origin of type IIB multi-Janus solutions, with C. Mascherpa, A. M. Uranga

TALKS

INVITED TALKS AND SEMINARS

May 2026 — *Anomalies at SUSY Interfaces of String Theory*, invited talk, Simons-Balseiro School and Workshop on Advanced Aspects of QFT, Instituto Balseiro, Bariloche

Jun 2025 — *Relative Quantum Gravity: Localized Gravity and the Swampland*, invited seminar, Max Planck Institute, Munich

Feb 2025 — *Relative Quantum Gravity: Localized Gravity and the Swampland*, invited seminar, Heidelberg University

Nov 2024 — *SymTFT Fans: The Symmetry Theory of 4d $\mathcal{N} = 4$ Super Yang–Mills on Spaces with Boundaries*, invited seminar, Universidad de Murcia

Apr 2024 — *Bubbles of Nothing: The Tunneling Potential Approach*, invited seminar, Tufts University

— *Dynamical Cobordism in AdS/CFT*, invited seminar, Northeastern University

OTHER TALKS AND POSTERS

Feb 2026 — *Lessons from Quantum Gravity and Holographic End-of-the-World Branes*, seminar, Instituto Balseiro, Bariloche

Jan 2025 — *Relative Quantum Gravity: Localized Gravity and the Swampland*, poster, Iberian Strings 2025, Bilbao

— *Relative Quantum Gravity: Localized Gravity and the Swampland*, seminar, Harvard University

Oct 2024 — *Topological Couplings in Holographic End-of-the-World Branes from String Theory*, String Pheno Seminars, online

Jun 2024 — *Dynamical Cobordism in AdS/CFT*, parallel talk, String Phenomenology 2024, Padova

— *Supersymmetric Boundary Conditions for the End of Spacetime*, parallel talk, SUSY24, IFT, Madrid

TALKS (CONTINUED)

- *Dynamical Cobordism in AdS/CFT*, poster, Strings 2024, CERN, Geneva

Mar 2024 » *Bubbles of Nothing: The Tunneling Potential Approach*, seminar, Harvard University

Jan 2024 » *Dynamical Cobordism in AdS/CFT*, String Pheno Seminars, online

Dec 2023 » *Dynamical Cobordism in AdS/CFT*, talk, 7th Winter Workshop, Universidad de Valencia

Dec 2022 » *Bubbles of Nothing: A Tunneling Potential Approach*, talk, Madrid Winter Workshop 2022, Universidad Complutense de Madrid

Jul 2022 » *Quantum Tunneling: From Bubbles of Nothing to String Theory Dynamical Cobordism*, parallel talk, String Phenomenology 2022, Liverpool

Mar 2022 » *At the End of the World: Local Dynamical Cobordism*, seminar, IFT, Madrid

STUDENT SUPERVISION

2026 » Co-advisor (with Gerardo Aldazábal) of Julián Flores' B.Sc. thesis, *String Theory and the Dark Dimension Scenario*, Instituto Balseiro

TEACHING

2024 – 2025 » Teaching Assistant in *Physics Laboratory*, Chemistry Degree at UAM, Madrid

2023 – 2024 » Teaching Assistant in *General Physics*, Telecommunications Engineering Degree at UAM, Madrid

- Teaching Assistant in *Quantum Mechanics I*, Physics Degree at UAM, Madrid

2022 – 2023 » Teaching Assistant in *Foundations of Physics III*, Physics Degree at UAM, Madrid

- Teaching Assistant in *Quantum Mechanics I*, Physics Degree at UAM, Madrid

2021 – 2022 » Teaching Assistant in *Classical Mechanics*, Physics Degree at UAM, Madrid

- Teaching Assistant in *Physics Laboratory*, Chemistry Degree at UAM, Madrid

REFEREEING

» Referee for Journal of High Energy Physics since May 2023

OUTREACH

2025 – 2026 » Advisor of Leo Corpas Martínez's final high school essay, *About the Possibility of Time Travel*

Sep 2023 » Outreach video in Spanish on why mathematics works, IFT YouTube channel, over 50k views: *¿Por qué funcionan las matemáticas?*

- Helped at "Escape-road: In search of Nobel and non-Nobel women scientists", an event for schoolchildren at the European Researchers' Night

May 2023 » Invited speaker at "Tarde de Física", an advanced outreach session for undergraduate physics students

Mar 2023 » Three-minute outreach video on my research on bubbles of nothing: YouTube

2021 – 2025 » High school outreach talks on black holes, cosmology, the history of the universe, astronomy, and quantum computers